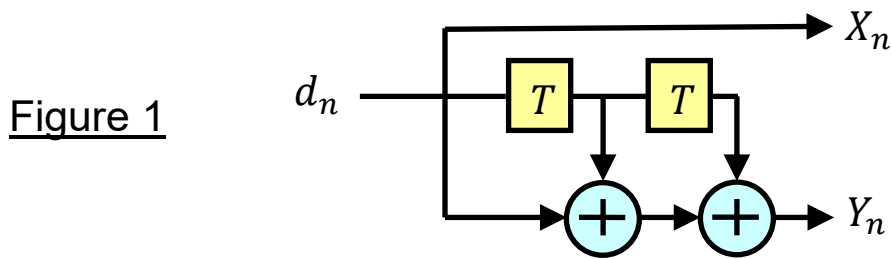


Problem-Solving Exercises

Coding Theory

Question 1

Consider the convolutional encoder depicted in Figure 1. At each time $n \cdot T$, where T and n denote the encoder period and an integer, respectively, the encoder is fed with an information bit d_n and generates two coded bits X_n and Y_n .



(a) Draw the trellis diagram for this encoder. **[10 marks]**

(b) Use this trellis diagram to determine the free distance, d_{free} , for this encoder.

Show the path(s) in the trellis that correspond(s) to this free distance.

[10 marks]

(c) Find the expression of the bit error probability obtained at high SNR using this convolutional code over a BPSK, AWGN channel.

What is the value of the asymptotic coding gain over uncoded BPSK?

[5 marks]

- (d) Do you think that the convolutional code studied in this question is optimal in terms of error performance?

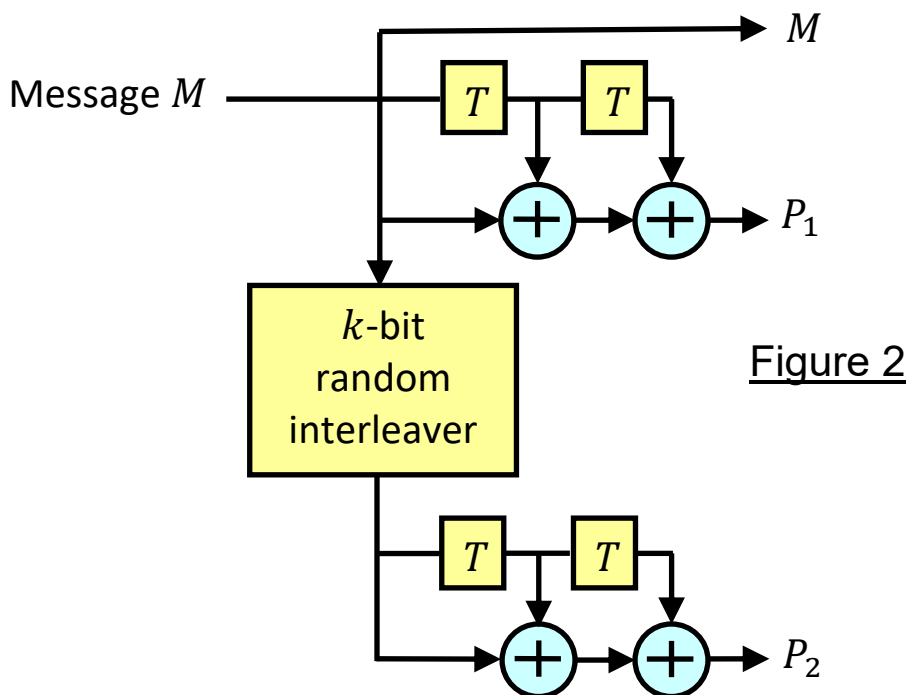
Or, do you think that a better error performance can be achieved by employing another convolutional code with identical constraint length and coding rate?

[5 marks]

- (e) We now use the convolutional encoder depicted in Figure 1 to design a turbo encoder, as shown in Figure 2.

The turbo encoder input is a k -bit message M . This message is encoded into three k -bit binary words. The first one is the message M itself.

The second binary word is the sequence P_1 of parity bits produced by the first convolutional encoder. Finally, the third binary word is the sequence P_2 of parity bits produced by the second convolutional encoder which is fed by an interleaved version of M .



Find the expression of the first term of the union bound for this turbo code.

To save time and space on your answer sheet, you are allowed to use some results given in the lecture notes, without having to demonstrate them once again.

[10 marks]

- (f) The turbo encoder studied in (e) is designed using convolutional encoders that are non-recursive.

Given the results obtained in (e), would you recommend using non-recursive convolutional codes for the design of turbo codes?

[10 marks]

Question 2

Consider the linear block code defined by the generator matrix

$$G = \begin{bmatrix} 100110 \\ 010101 \\ 001011 \end{bmatrix}.$$

- (a) Determine the minimum distance, d_{min} , between codewords for this code.

[10 marks]

- (b) Find the expression of the bit error probability obtained at high SNR with this code. Assume transmission over a BPSK, AWGN channel and the use of maximum-likelihood (ML) decoding.

What is the value of the asymptotic coding gain over uncoded BPSK?

[5 marks]

(c) Find the parity-check matrix H for this code. **[10 marks]**

(d) Draw the Tanner graph for this code. **[5 marks]**

(e) Assume transmission over a binary symmetric channel (BSC), and that the received word is $R = (111110)$.

Use the bit-flipping algorithm to decode R .

Comment on your result.

[10 marks]

(f) Modify the Tanner graph determined in (d) in order to improve the error-correction capability of the code when bit-flipping decoding is used.

As an illustration, use the modified Tanner graph to decode the received word $R = (111110)$.

Comment on your result.

[10 marks]

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